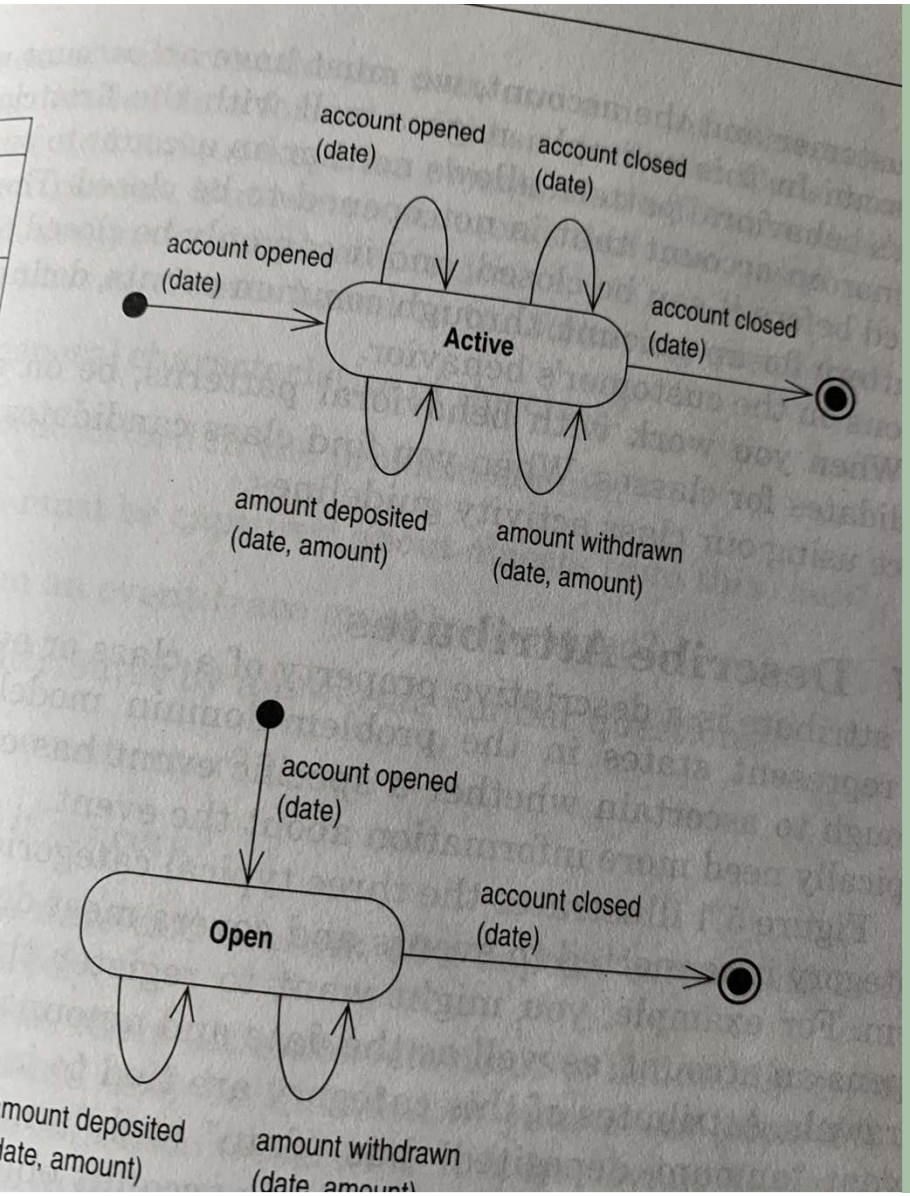


System Development

Session 3 – Problem Domain: Behaviour

Anders Bruun

Previous Session

Classes Activity

Selcting elements of a Problem Domain model

Object (look for nouns)
An entity with identity, behaviour and state

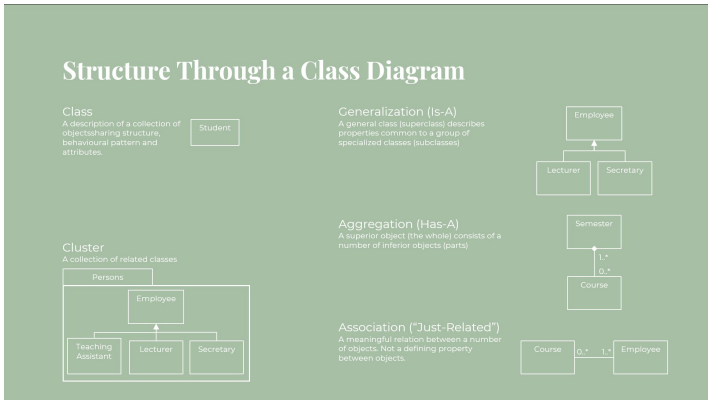
Class (look for nouns)
Collection of objects sharing structure, bahavioural pattern and attributes

Event (look for verbs)
Instantanious incident involving one or more objects

Events Classes

	Student	Lecturer	Course	...
Signed up	x			
Employed		x		
Exam planned		x	x	
Created			x	
...				

Classes and Events

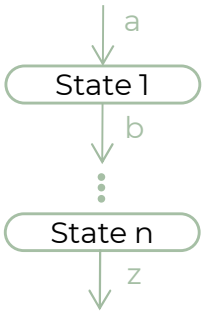


Structure

Behavioural Patterns - State Charts

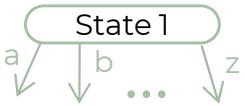
Sequence

Events in a set occur one by one



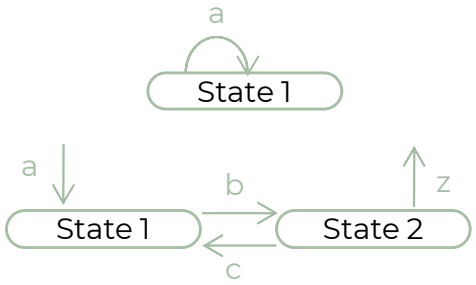
Selection

Exactly one out of a set of events occur



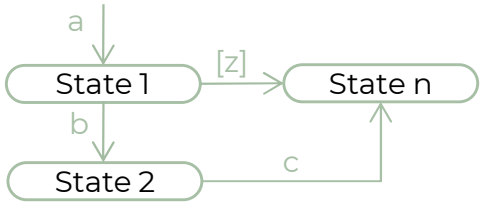
Iteration

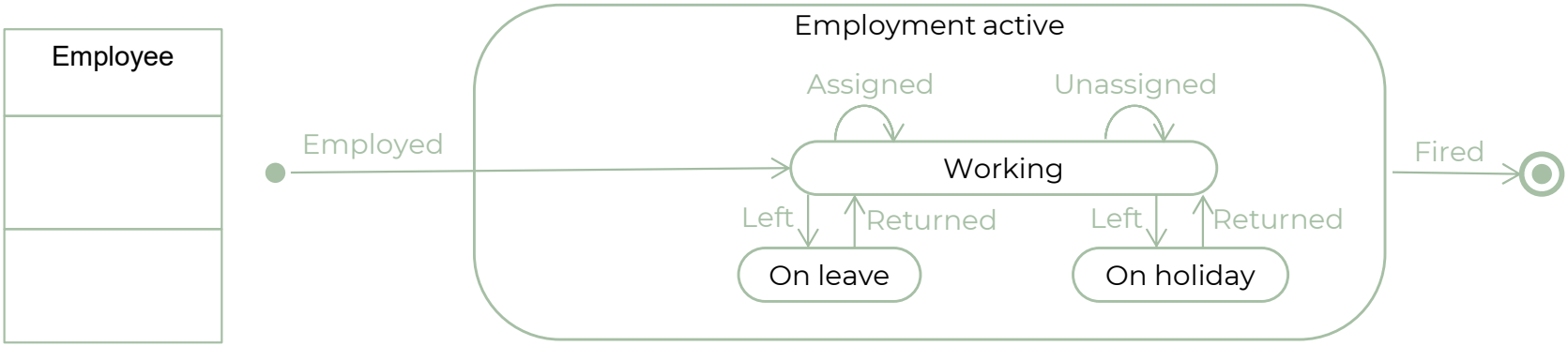
An event occurs zero or more times



Conditional

Transition happens when a stated condition becomes true




Employee
Employment active
Assigned
Unassigned
Employed
Working
Fired
Left
Returned
Left
Returned
On leave
On holiday

An Example

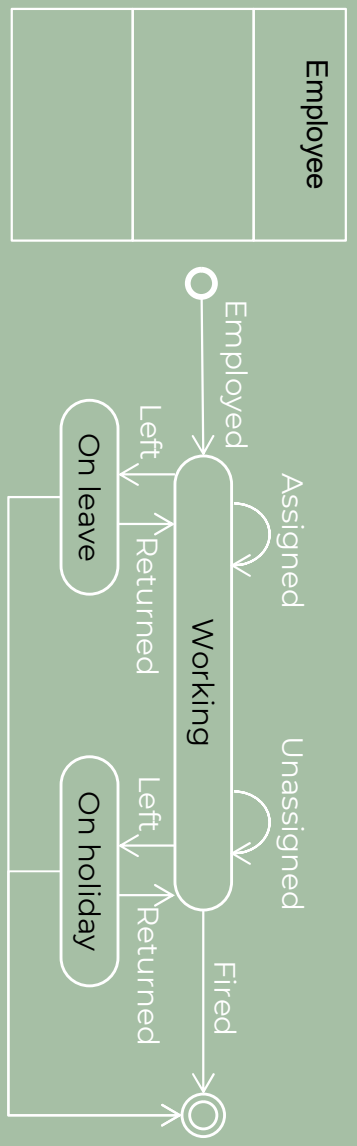
Tips and Tricks

... for describing **behavioural patterns** for a class:

- Events as the point of departure
- Identify first and last events in an objects lifespan
- Don’t go for ”Create” and ”Delete” (system events)

Also, ask yourselves:

- Is the overall form structured or unstructured?
- Which events occur in a sequence?
- Are there any alternative events?
- Can a given event occur more than once?



ations that lead back to the "Open" state. In this way, we arrive at the stat
echart diagram illustrated in Figure 5.1.

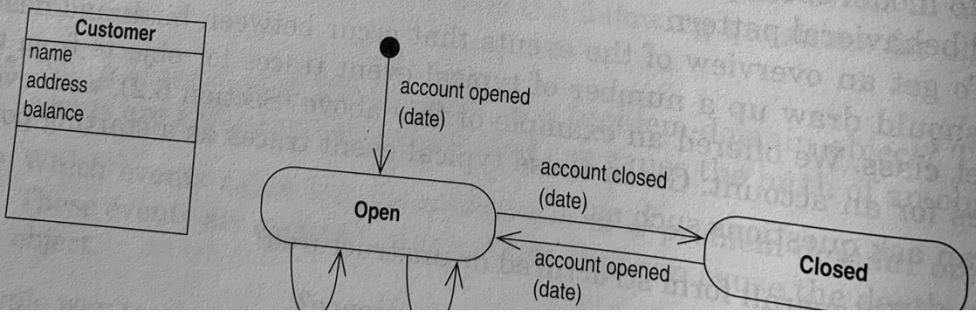
Sufficient, but Simple

When we make behavioral patterns, we have the following—sometimes conflicting—goals:

- The behavioral pattern should be sufficiently precise to describe all legal, and thus all illegal, event traces.
- The behavioral pattern should provide an overview and thus be as simple as possible.

We can usually avoid the potential conflict by excluding very specialized and detailed aspects of the behavior pattern from the diagram. Otherwise, we can end up with very complicated diagrams. When necessary, you can describe typical behavior in a simple diagram and supplement it with a comment about the specialized aspects.

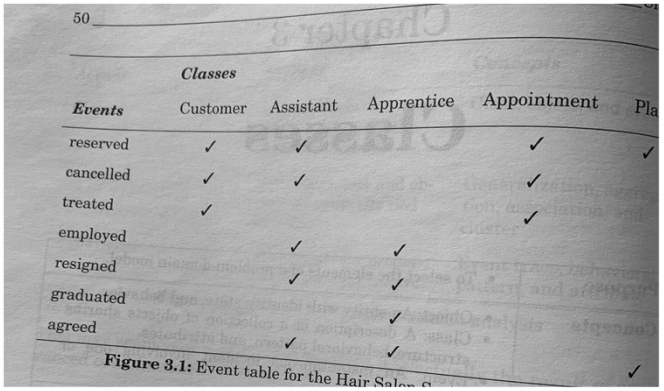
Our first goal is to produce a behavioral pattern that is sufficient to capture all allowable event traces. To accomplish this, we begin by describing a typical object's behavior. Later, we can integrate more unusual event traces to the description.



"Sufficient, but simple."

—Mathiassen et al., 2000.

Update the Event Table

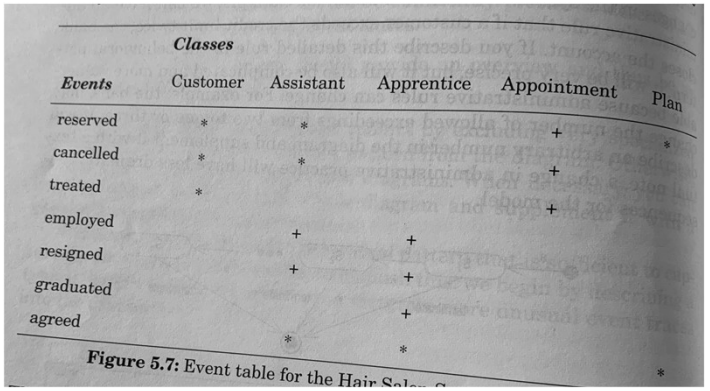


50

Events	*Classes* Customer	Assistant	Apprentice	Appointment	Pla
reserved	✓	✓		✓	✓
cancelled	✓	✓		✓	
treated	✓			✓	
employed		✓	✓		
resigned		✓	✓		
graduated			✓		
agreed		✓	✓		

Figure 3.1: Event table for the Hair Salon S

From this...



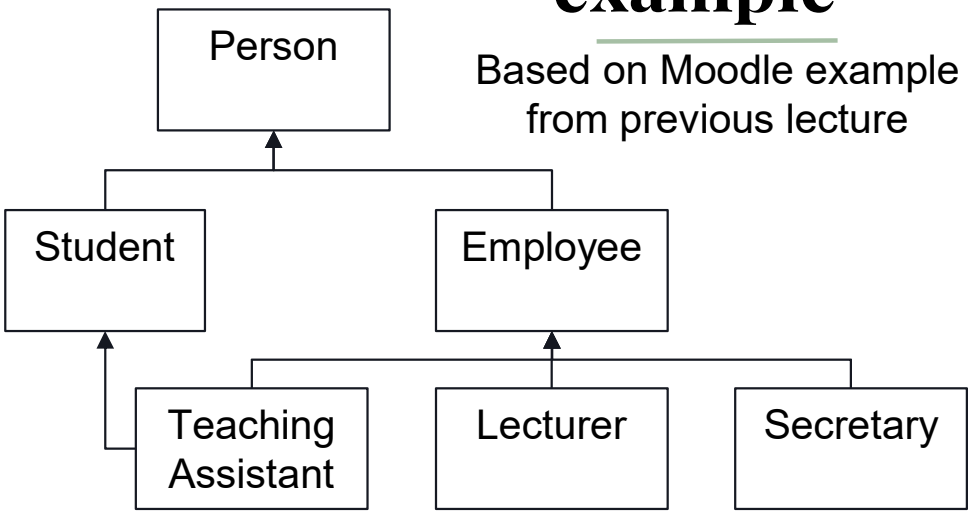
Events	*Classes* Customer	Assistant	Apprentice	Appointment	Plan
reserved	*	*		+	*
cancelled	*	*		+	
treated	*			+	
employed		+	+		
resigned		+	+		
graduated			+		
agreed		*	*		*

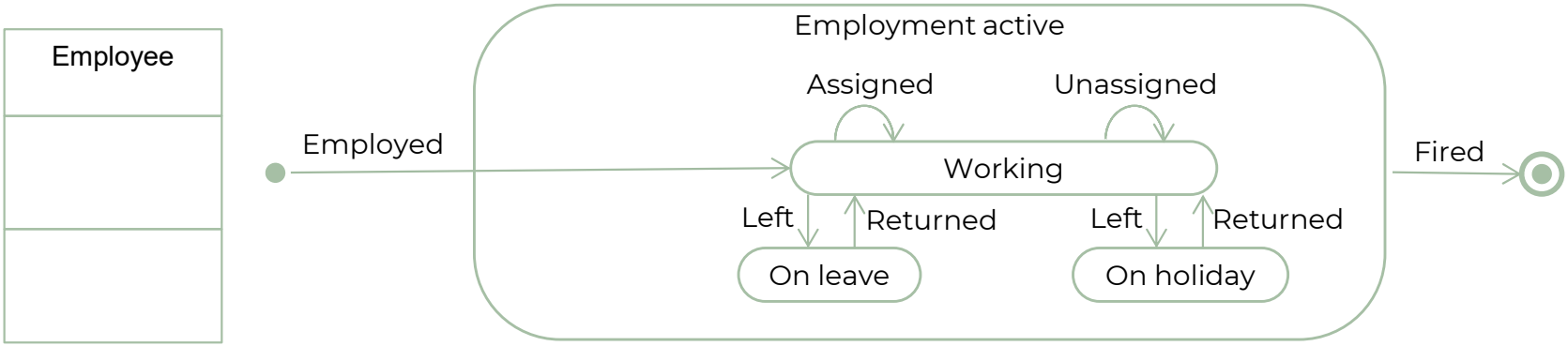
Figure 5.7: Event table for the Hair Salon S

... to this

Behaviour example

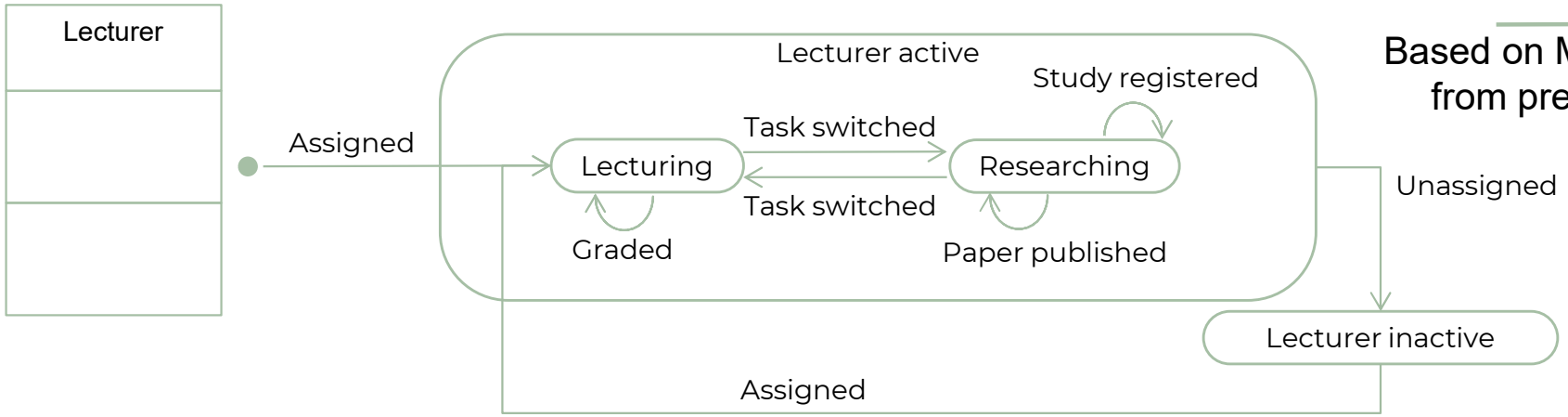
Based on Moodle example from previous lecture





Behaviour example

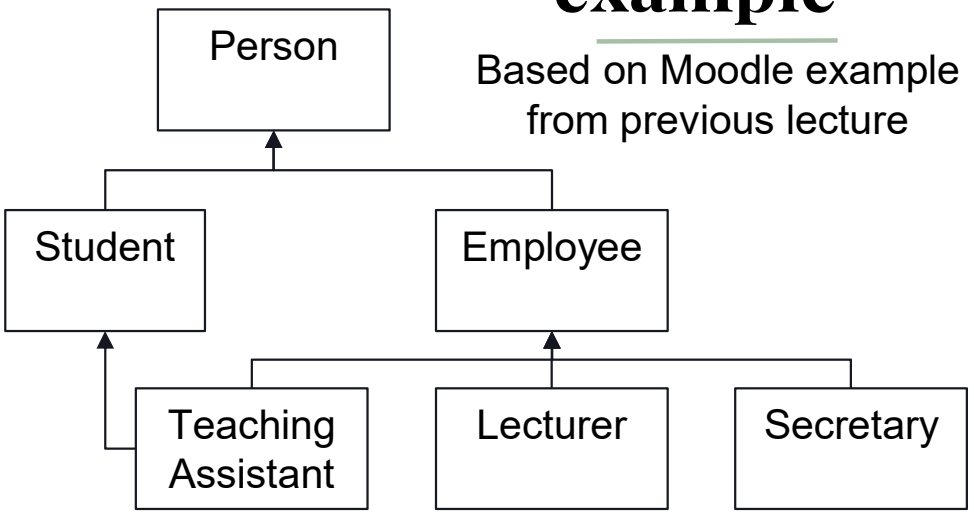
Based on Moodle example from previous lecture



Events	Classes			
	Employee	Lecturer	Student	Person
Signed up			+	
Graduated			+	
Employed	+	+		
Assigned	*	*	*	*
Fired	+	+		
Unassigned	*	*	*	*
Graded		*	*	
Course completed			*	
Left	*	*	*	*
Returned	*	*	*	*
Task switched		*		
Study registered		*		
Paper published		*		

Behaviour example

Based on Moodle example from previous lecture



Attributes

Descriptive properties of a class or an event

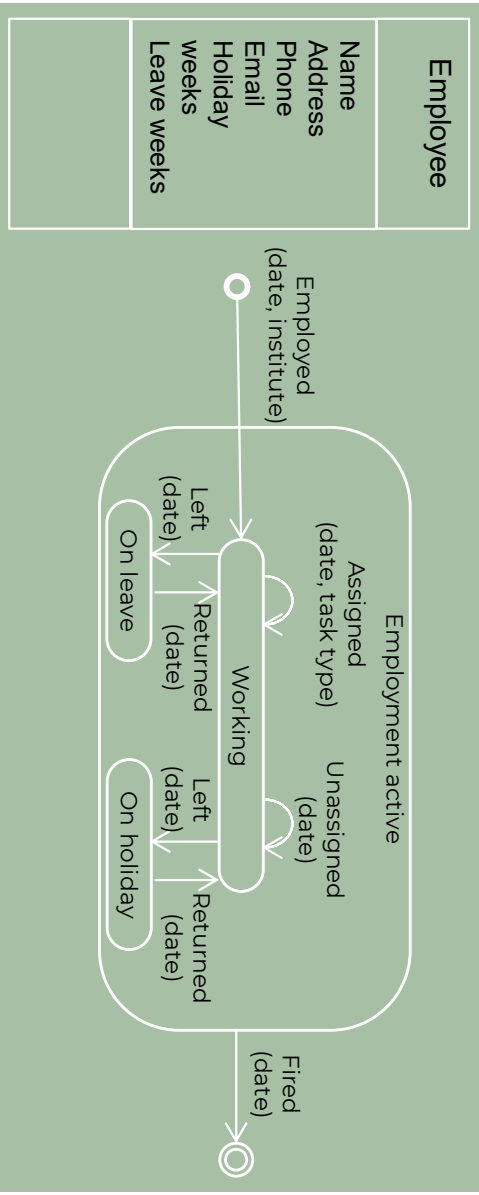
Identifying Attributes for a Class

Three types of attributes:

- Information connected to single events
- General information connected to objects as a whole
- Derived information based on other attribute values

Ask yourselves:

- What are the general characteristics of the class?
- How is the class described in the Problem Domain?
- What basic data must be captured from this class?
- What results from events must be captured?
- What states can the object be in?



Behaviour of Objects in PD

Exercise 3.1

Choose 1-3 central classes from the F-klubben case (assignment 2.1 from the previous session). Create UML state-transition diagrams to describe the behavioural patterns for objects within each of these 1-3 classes.

Think in terms of defining the various states that objects can be in during their "lifespan" and the events causing transitions between states, e.g. "birth" and "death" events and every state and event that occur between these events.

Make sure there is correspondance between events listed in the event table from exercise 2.1 and your state-transition diagrams and consider the assessment criteria from the book (the "Sufficient, but simple" tips and tricks slides).

PD Object Attributes

Exercise 3.2

An attribute is a descriptive property of a class or an event. A class typically consists of multiple attributes, classical attributes are e.g. “name”, “address” etc.

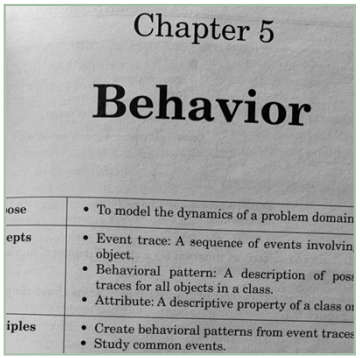
Identify attributes for the 1-3 central classes that you selected in exercise 3.1. Use a maximum of two words to describe each attribute. Use the questions suitable for identifying attributes as defined in the course material (book and slides).

Behaviour and Attributes for your projects

Exercise 3.3

Identify and describe behavioural patterns and attributes for all the classes that you have identified related to the Problem Domain model of your semester project cases, following the steps of exercises 3.1 and 3.2.

Until we meet again


Chapter 5

Behavior

ose	• To model the dynamics of a problem domain
epts	• Event trace: A sequence of events involvin object. • Behavioral pattern: A description of poss traces for all objects in a class. • Attribute: A descriptive property of a class o
iples	• Create behavioral patterns from event traces • Study common events.

